

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

SP9 - Enterprise Risk Management

Total Marks: 100

Indicative Solution

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable

Solution 1:

(i) The following outlined points may be included in the response to the CEO regarding the importance of ERM and how it compares with conventional risk management-

- ERM offers a holistic approach, integrating risk management across all departments and functions, providing a comprehensive view of risks at the enterprise level.
- ERM ensures compliance with regulatory requirements and best practices.
- Conventional risk management is often reactive, focusing on mitigating risks as they arise. The approach can be more about damage control after risks have materialized.
- ERM emphasizes the proactive identification of potential risks and opportunities, anticipating issues before they become significant problems.
- ERM ensures that the senior management is more informed
- The company's risk exposure may be better understood with ERM than with conventional risk management, conventional risk may involves more silo-based approach with each department or function managing its risks independently, leading to a fragmented view of the company's risk profile.
- Risk return and growth can be linked more precisely with ERM
- External factors' impact can be accounted for, and they become more actionable
- Risk-strategy may be defined with more relevance and aligned with risk tolerance and appetite
- Centralised ERM helps coordination better across the company
- Risk-information sharing becomes more convenient via the ERM approach
- ERM may help optimize management's time and resources and may help allocate capital more efficiently
- The pricing of the company's services, and products gets more rationalized with ERM
- Reactions to hazardous events become more prompt, and losses may be minimized more effectively with ERM.
- ERM manages risks better and optimizes risk transfers
- The recent event also calls for an integrated and holistic risk-management approach with ERM than a conventional approach.

(Max 7)

(ii) - Market risk

- Interest rate risk: the fluctuations in interest rates Up or Down affecting the investment yields of the company. This may lead to insufficient returns to its customers
 - Equity risk: the equity returns' fluctuations affecting the returns. It depends on the extent of investment in equity and the volatility in such returns.
 - This may also be heightened if integration post-acquisition would have changed the market risk sensitivity of the company as a whole.
- Liquidity risk: the inability to meet expected and unexpected obligations without additional cost. This may also involve the start-up's liabilities and payables.
- Credit risk: the risk that the counterparty will be unwilling or unable to make the required payments, also from the start-up's counterparties.
- Operation risk: inadequate or failed processes, people, and systems in the company. It could be related to IT systems, financial fraud, cyber incidents, etc. Start-ups' systems may cause more adverse system errors, faults, inefficient integration, etc.
- Technological risk/data breaches- The integration may have exposed sensitive customer and business data to unauthorized access, increasing the risk of data breaches and associated penalties.
- Regulatory risk: risk due to changes in regulatory provisions of investments etc. Regulator's view on merger/acquisition.
- Regulatory risk (post)- Higher scrutiny to understand data protection violations.
- Reputational risk: risk due to external or internal adverse events, such as cyber incidents, and risks that may have been added due to the start-up's integration.
- Project risk: due to the inefficiency and failure of a project or portfolio
- Strategic risk: due to inadequacy of returns from an investment and business strategy. The start-up's business strategy might also contribute to strategic incongruence and risks.
- Strategic Risk (post)- The alignment between the start-up's technology and the company's long-term strategy could be at risk if the integration is not managed effectively, potentially leading to a misalignment of business goals.
- Concentration risk: due to a lack of diversity in investments and risk exposures, which are also added by the start-up's investments.
- Systemic and contagion risk: due to interdependent entities within the system
- Identification and analysis of cyber risk:
 - Hacking, security breaches, data theft, cyber terrorism, etc.
 - Business interruption, reputational damage, legal liability

- The source and nature of cyber risk are changing and evolving, so it is difficult to manage and monitor.
- Incident management processes, strong IT security and clear policies are important.

(Max 10)

(iii)**(a)**

- Reporting of Information Security Head to the Head of IT is not ideal.
- There may be a conflict of interest with this reporting.
- The development and operations of IT which is handled by Head IT should be segregated from the IT security to avoid the conflict of interest and have best practice.
- The IT security should have independence and should ideally be reporting to the CRO.
- The respective teams should also have separation of duties.

(Max 2)

(b) Recommendations:

- Form a cyber crisis management team
- Define and constitute a cyber incident escalation matrix
- Conduct regularly cyber awareness sessions and training sessions
- Conduct mock drills of disaster recovery and table top exercises
- Design and implement IT security policies and procedures
- Have the time and point recovery based on criticality of applications, etc.

(Max 2)

(iv) (a)**Measurement-
Credit Risk:**

- Value at risk methods may be used
- A 1% or 5% VaR for 1 year period may be relevant for the company
- Additionally the 1% or 5% Tail VaR may also be utilised
- Concentration of risk may also be assessed
- Sector exposure of the investments should be monitored
- The relevant clusters of investments and the effect of higher than expected default experience should be investigated

Market Risk:

- Duration mismatch between asset and liability
- Currency risk if any
- Equity market risk – fall in equity value in equity market
- VaR or Tail VaR analysis of asset and liability portfolio should be conducted

(4)

(b)

Credit Risk:

Limits may be set as:

- Standalone VaR of portfolio
- Exposure wise counterparty risks
- Maximum exposure to a sector

Market risk:

Limits may be set as:

- Embedded options if any should be considered and exercise to considered for placing limits
- VaR limits may be placed
- Stochastic simulations may be carried out if feasible to decide extent of limits
- Deterministic approaches to tests the ALM mis(match) should be investigated and match subsequently and reasonably

(Max 3)

(c)

Credit Risk:

If limits are breached:

- Diversify the exposure to the risky sectors
- Dilute the concentration of such default experience areas
- Other instruments such as credit default swaps may be tried
- Credit models may be used to quantify the losses

Market Risk:

If limits are breached:

- Improve ALM matching
- Adjust size of equity and interest-rate sensitive portfolio as per the stress tests' outcomes
- Derivative instruments may be considered to hedge the positions suitably
- Currency risk if any may be considered with futures or forwards etc.

(4)

(v)

<i>Stakeholder</i>	<i>Perspectives</i>
- Board of Directors	Compliance with the relevant regulation and perform duty of care
- Shareholders	To achieve risk-adjusted returns
- Employees	Growth of benefits and employment security
- Customers	To achieve value in the investments
- Credit rating agencies	To avoid conflict of interest and reputation risk
- Auditors	To avoid risk of litigation and reputational risk
- Government/Regulators	To prevent contagion risk, regulatory arbitrage, Market stability, protection of stakeholders

(Max 7)

[39]**Solution 2:**

- (i) (a) A risk measure is a statistical indicator of the amount of risk a particular portfolio or set of assets is exposed with a certain loss distribution and time horizon. (1)

(b) Time horizon for risk measure is decided based on:

- Time to recover from a loss event
- Time to reinstate risk management
- The regulatory prescription if any
- Risk policy provision, etc.
- Investor's views and preferences

(2)

(c) VaR is the maximum potential loss which is not exceeded with a given high probability over a given time period.

Tail VaR - the expected loss given that a loss over the specified VaR has occurred

(2)

(d)

Tail-VaR:

Adv.:

- Considers losses beyond VaR
- It's a coherent risk measure

Dis-adv.:

- Distribution choice is not easy or sometimes feasible
- Assumption sensitive

(2)

(ii) Each asset's standard deviation = $0.02 \times 150000 = 3000$
 Portfolio variance = $2 \times 3000^2 + 2 \times 3000^2 \times 0.37 = 24660000$
 5-year std.dev. = $\sqrt{5 \times 24660000} = 11104.05$
 Assuming returns $X \sim N(\mu, \sigma^2)$
 $\mu = 0$. This has limitation as the returns may not be *iid* and μ may not be zero in reality.
 $N(-1.88) = 0.03$
 So, 5-year VaR = $1.88 \times 11104.05 = 20875.62$ (5)

(iii) Reasons to consider development of a credit risk model:

- Credit risk is a key risk for the insurer to back the long-term liabilities
- Tail losses in credit default can be important to analyse
- The portfolio allocation in the investment sectors may be reflected due to this modelling exercise
- A simplistic model may miss the insights and may not stand the actual outcomes than estimated
- Risk-decisions may be better facilitated by such credit modelling exercise
- Model can incorporate stress and scenarios tests
- Risk limits can be set out of this
- Insurer's business projections may be better arrived than now after the model development
- Risk-taking and upside potential can be explored
- Regulatory acceptance may improve due to such an approach
- Insights for other risks may also be arrived better
- insurer can choose and diversify and adjust the portfolio based on the model outcomes

(6)

(iv) (a) Possible scenarios:

- 76% probability that loss in neither occurs, loss = 0
- 19% probability that loss in only A occurs, loss of INR 99Cr
- 4% probability that loss in only B occurs, loss of INR 100Cr
- 1% probability that loss in both A and B occur, loss of INR 199Cr
- 99% VaR = INR 100Cr

(6)

(b) Considerations and other points:

- Adjustments for correlation and dependency required.
- These correlations required in stress times

- Tail dependency to be considered
- Efficient methods of aggregation of capital to be considered
- Diversification benefits to be included
- Purpose of allocation and stakeholders to be considered
- Element of judgment in allocation is likely to remain
- Allocation:
 - Can be centrally retained
 - Lack of understanding
 - Risk measure may be used
 - Proportional spread may be used
 - Marginal approaches may be used
 - Adheres to the finance principle of marginal pricing
 - Stand-alone basis for capital allocation to be considered
 - Game theory may be used
 - Shapley methods references to the average of marginal additional
 - Euler method
 - May be computationally intensive
 - Pro-rata approach:
 - may be references to some basis – premiums, reserves etc.

(9)
[33]

Solution 3:

- (i)
- Bottoms-up models and statistical analysis could have been employed to analyse historical data and identify potential anomalies or irregularities in financial transactions.
 - Utilization of Risk Analytics: company could have utilized risk modelling techniques to detect patterns indicative of operational risks, not just in financial transactions but also in technological operations.
 - Rigorous Scenario Analysis: This approach could have involved assessing the impact of competitive pressures and financial irregularities not only concerning other market players but also considering how technological vulnerabilities may exacerbate these risks.
 - Assuming sufficient market data, estimating potential financial losses attributable to both financial and technological operational vulnerabilities could have provided a more comprehensive view of risk exposure.

- Probabilistic Risk Assessment: While utilizing simulations to assess risk probabilities, company could have incorporated data related to technological events and vulnerabilities if available, enhancing the accuracy of risk assessment.
- Value-at-Risk Calculation: Estimating the maximum potential loss with VaR analysis should have considered risks arising from both financial misconduct and technological failures.
- Extreme Value Theory Application: Assessing tail risks associated with operational vulnerabilities should have also included considerations of technological risks, such as cyber threats or system failures.
- Long-Term Risk Trend Identification: Analysing trends to anticipate evolving risks should have encompassed trends in technological advancements and vulnerabilities as well.
- Correlation Analysis: Assessing interdependencies between risk factors should have included correlations between financial misconduct and technological vulnerabilities.
- Sensitivity Analysis: Evaluating financial performance under various risk scenarios should have accounted for potential technological disruptions and their impact.
- Loss Severity Estimation: Employing actuarial methods to estimate potential loss severity should have considered technological risks alongside financial risks.
- Visualization through Risk Heat Maps: Providing a clear depiction of risk distribution and severity should have included both financial and technological risk factors.
- Historical Loss Data Analysis: Identifying patterns to anticipate future risks should have incorporated historical data related to both financial misconduct and technological failures.
- Monitoring Key Risk Indicators: Detecting early warning signs of operational risk deterioration should have included indicators related to both financial irregularities and technological vulnerabilities.
- Business Process Mapping: Identifying vulnerabilities and control weaknesses should have encompassed both financial processes and technological infrastructure.
- Regulatory Compliance Metrics Tracking: Monitoring compliance with regulatory provisions should have included adherence to regulations concerning both financial reporting and technological security.
- Root Cause Analysis: Identifying underlying factors contributing to operational risk events should have explored root causes related to both financial misconduct and technological failures.

- Cultivating Continuous Improvement: Establishing a culture of learning and adaptation to enhance risk management practices should have encouraged innovation and improvement in both financial and technological risk management processes.

(7)

(ii) The company could have handled the given risks as follows:

- The company should have decided formally as policy matter on the risk exposure which it should have.
- It should have assigned the probability, impact and capital for such risks
- These risks should have been integrated by the company as part of its overall risk framework
- Limits on overall operational-risks should have been placed
- Sub-limits on overall (sub)-operational-risks should have been placed
- Internal controls such as internal audits, sampling tests of records should have been in place
- Mock drills and business continuity planning w.r.t. the external market should have been conducted.
- Alternate vendors/providers/system quality analysis should have been conducted regularly
- Comprehensive system audit should have been conducted on IT systems
- Corresponding gaps should have been addressed
- The company being large company should have incorporated the possible reputational risk scenarios and actions taken thereof
- Internal controls/access controls should have been in-place
- Independent checks and balances at every node in the supply chain should have been there
- Risk retention in terms of self-insuring could have reduced the impact
- Risk management actions and the accountability governance structure should have been made clear
- Suitable liability or other form of insurance could have been taken
- Loss indicators for Op-risk should have been prepared
- Op-risk broad definition and external warning indicators might have helped mitigate the risks for the company.
- Controls – qualitative and quantitative in-place could have been helped

- Compliance function and regular contact with regulator may have helped the warnings and early signals
- Outsourcing could have been considered for those technological operations in question
- Crisis management planning and systems due diligence should have been in place

(11)

- (iii)
- Identify and prioritize risks related to company's pension fund stability, employee morale amidst financial turmoil, and safeguarding company's reputation.
 - Assess the impact of company's financial distress through projections and stress testing on pension fund viability and employee retirement benefits.
 - Optimize company's pension fund investment portfolio to mitigate market volatility in light of the company's financial challenges.
 - Strengthen oversight mechanisms within company to ensure prudent management of pension funds and rebuild trust with employees through employee connect sessions.
 - Communicate transparently with company's employees about the status of their retirement savings to mitigate anxiety and uncertainty.
 - Develop contingency plans to respond effectively to potential pension fund funding shortfalls or liquidity crises.
 - Consider environmental, social, and governance (ESG) factors in company's investment decisions to align with responsible investing practices which may help improve the public perception.
 - Continuously monitor performance indicators within company's pension fund management and adjust strategies accordingly.
 - Ensure company's compliance with pension regulations and reporting requirements to mitigate legal and regulatory risks.
 - Explore options like purchasing annuities for company's pension benefits to provide stability for retirees.
 - Conduct thorough due diligence on administrators and service providers to mitigate operational risks within company's pension management.
 - Offer financial counselling and support services to employees impacted by pension fund uncertainties.
 - Regularly assess company's pension liabilities and funding requirements in light of the company's financial situation.

- Forecast company's pension obligations and assess the sustainability of funding given the current financial challenges.
- Quantify the financial impacts of company's financial distress on pension fund solvency and capital adequacy.
- Utilize annuitant's mortality tables and accurate actuarial calculations to manage risks associated with company's pension liabilities.
- Align company's pension fund assets with liabilities to reduce volatility and ensure long-term sustainability.
- Adjust pension contribution levels within the company based on the company's financial health and funding status changes.
- Establish reserves within the company to protect pension funds against potential funding shortfalls or economic downturns.
- Conduct sensitivity analysis to assess the impact of key assumptions on company's pension fund outcomes, considering the company's current financial condition.

(10)

[28]
